

How the Proof Works

Thomas Birnie

The three-dimensional Navier–Stokes regularity problem asks whether a smooth solution can lose its smoothness in finite time. Fix a smooth, divergence-free datum u_0 and a viscosity $\nu > 0$.

Take a vortex tube as the physical picture. As it stretches and thins, the fluid around its shrinking core turns faster and faster, and the velocity changes across a smaller distance. The tube cannot lengthen without becoming narrower.

To see where that concentration sits in the Sobolev scale, rescale the fluid:

$$u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t), \quad \|u_\lambda(t)\|_{\dot{H}^s} = \lambda^{s-1/2} \|u(\lambda^2 t)\|_{\dot{H}^s}.$$

Below $s = \frac{1}{2}$, concentration makes the Sobolev norm smaller. Above $s = \frac{1}{2}$, it makes the norm larger. At $s = \frac{1}{2}$, the power of λ disappears. The exponent vanishes between the energy level below it and the first-derivative level above it. That unchanged middle level is the critical height. At this height, viscosity and nonlinear transport have equal scaling weight.

The kinetic energy can remain finite while smaller and smaller regions carry steeper gradients.

In order for a singularity to form, velocity has to blow up. In order for that to happen, its acceleration would also have to blow up, which means its jerk has to blow up, and so on, and so on:

$$u, \quad \partial_t u, \quad \partial_t^2 u, \quad \dots$$

If a singularity were forming, what we would expect to see as we move up those time derivatives is less spatial regularity, not more.

What the time derivatives reveal

Write out the first three derivatives:

$$\begin{aligned}\partial_t u &= \nu \Delta u - (u \cdot \nabla)u - \nabla p, \\ \partial_t^2 u &= \nu \Delta \partial_t u - (\partial_t u \cdot \nabla)u - (u \cdot \nabla)\partial_t u - \nabla \partial_t p, \\ \partial_t^3 u &= \nu \Delta \partial_t^2 u - (\partial_t^2 u \cdot \nabla)u - 2(\partial_t u \cdot \nabla)\partial_t u \\ &\quad - (u \cdot \nabla)\partial_t^2 u - \nabla \partial_t^2 p.\end{aligned}$$

By the third line, the nonlinear term is visibly branching. The binomial form writes the complete expansion at once:

$$\begin{aligned}\partial_t^{r+1} u + \sum_{a=0}^r \binom{r}{a} ((\partial_t^a u) \cdot \nabla) \partial_t^{r-a} u + \nabla \partial_t^r p \\ = \nu \Delta \partial_t^r u.\end{aligned}$$

Set

$$V_r := \partial_t^r u, \quad Q_r := \partial_t^r p.$$

Now ask what happened to spatial regularity while those time derivatives were climbing. Write $\Lambda := (-\Delta)^{1/2}$. At time-derivative order r and spatial height s , measure V_r by

$$E_{r,s}(t) := \frac{1}{2} \|\Lambda^s V_r(t)\|_2^2.$$

Call the complete transport and pressure contribution $\mathcal{P}_{r,s}$:

$$\mathcal{P}_{r,s} := -\operatorname{Re} \left\langle \Lambda^s V_r, \Lambda^s \left[\sum_{a=0}^r \binom{r}{a} (V_a \cdot \nabla) V_{r-a} + \nabla Q_r \right] \right\rangle_{L^2}.$$

Taking the H^s energy of the differentiated equation gives

$$E'_{r,s} = \mathcal{P}_{r,s} - 2\nu E_{r,s+1}.$$

This is where viscosity enters the argument. The time derivative of the energy at height s contains the energy at height $s+1$, with coefficient 2ν . Start with the original velocity at the critical height, $H := E_{0,1/2}$. Then

$$H' = \mathcal{P}_{0,1/2} - 2\nu E_{0,3/2}.$$

Each derivative of H is built from the temporal responses above. Differentiating $H = \frac{1}{2}\|\Lambda^{1/2}u\|_2^2$ distributes the time derivatives between its two copies of u :

$$H^{(n)}(t) = \frac{1}{2} \sum_{a=0}^n \binom{n}{a} \operatorname{Re} \left\langle \Lambda^{1/2} V_a(t), \Lambda^{1/2} V_{n-a}(t) \right\rangle_{L^2}.$$

Iterating the viscous energy identity $n \geq 1$ times gives

$$\frac{d^n H}{dt^n} = (-2\nu)^n E_{0,n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{0,j+1/2}.$$

The top term is the Sobolev energy at height $n + \frac{1}{2}$. As temporal order rises, Sobolev height rises with it. That is the reversal. The singularity picture led us to expect spatial regularity to fall away as we followed velocity, acceleration, and jerk. The equation exposes higher spatial energy instead.

In three dimensions, the Sobolev embedding theorem gives

$$s > k + \frac{3}{2} \implies H^s(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3), \quad \bigcap_{m < \infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

This is the payoff waiting at the endpoint. If every finite height reaches T_* , the endpoint lies in H^∞ and is smooth.

How the heights reach the endpoint

The calculation has exposed the heights. The finite rectangles carry them to T_* .

The index r tells us which time derivative V_r we are following, and the index m tells us the Sobolev height at which it is measured. The pairs (r, m) form the mixed grid. Cutting that grid off at a finite temporal depth N and a finite spatial height M gives one rectangle.

For each $n \leq N$, the adjacent pair

$$V_n \in L_t^2 H_x^{M+1}, \quad V_{n+1} = \partial_t V_n \in L_t^2 H_x^{M-1} \implies V_n \in C_t H_x^M$$

carries V_n continuously through T_* at the middle height H^M . Collecting finitely many adjacent pairs, together with their pressure rows Q_n , gives the pressure-complete rectangle $\mathcal{R}_{N,M}[u_0]$.

The rectangle estimate keeps every fixed finite block controlled through T_* . Larger blocks contain smaller ones, and their shared coordinates agree because every block comes from the fluid generated by u_0 . Taking all finite blocks together gives $\mathcal{I}[u_0]$, whose velocity belongs to

$$\bigcap_{r,m<\infty} H^r(0, T_*; H^m(\mathbb{R}^3)).$$

Every finite temporal order and every finite spatial height are now present at once.

Return to the zeroth row $V_0 = u$, denoted $\mathcal{S}_0 := \pi_0 \mathcal{I}[u_0]$. For every finite m , the intersection gives

$$u \in H^1(0, T_*; H^m(\mathbb{R}^3)) \hookrightarrow C([0, T_*]; H^m(\mathbb{R}^3)), \quad m < \infty.$$

The original velocity now has one endpoint in all of those spaces:

$$u_* := u(T_*) \in \bigcap_{m<\infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

The singularity picture led us to expect less spatial regularity as the time derivatives rose. The intersection reaches T_* with every finite spatial height intact.

Local smooth existence from u_* continues the solution through $[T_*, T_* + \delta]$. Uniqueness makes that continuation agree with the solution on $[0, T_*)$. This crosses a supposedly maximal finite time, so

$$T_* = \infty.$$

Where the estimates come from

Now choose any classical estimate. For finite r, k , $2 \leq p \leq \infty$, and

$$m > k + \frac{3}{2} - \frac{3}{p},$$

the intersection and Sobolev embedding give

$$\partial_t^r u \in C_t H_x^m \implies \nabla^k \partial_t^r u \in L_t^\infty L_x^p.$$

That one rule gives, for example,

$$\begin{aligned} \mathcal{I}[u_0] &\longrightarrow C_t H_x^1 \longrightarrow L_t^\infty L_x^6, \\ \mathcal{I}[u_0] &\longrightarrow C_t H_x^2 \longrightarrow L_t^\infty L_x^\infty, \\ \mathcal{I}[u_0] &\longrightarrow C_t H_x^3 \longrightarrow L_t^\infty W_x^{1,\infty}. \end{aligned}$$

The H^3 line gives, on any finite interval $[0, T]$,

$$\int_0^T \|\nabla u(t)\|_{L^\infty} dt \leq CT \sup_{0 \leq t \leq T} \|u(t)\|_{H^3} < \infty.$$

Change r , k , or p , and choose the corresponding finite height. The intersection already contains it.

The familiar continuation estimates are read from the intersection afterward. They are derived from u_0 through the intersection, not the other way around.

The one-line version is

$$\begin{aligned} u_0 &\longrightarrow \{(V_n, Q_n)\}_{n \geq 0} \longrightarrow \{\mathcal{R}_{N,M}[u_0]\}_{N,M < \infty} \longrightarrow \mathcal{I}[u_0] \\ &\longrightarrow \mathcal{S}_0 \longrightarrow u_* \in H^\infty \longrightarrow T_* = \infty. \end{aligned}$$